

grade

```
main.dart U X
1 import 'package:flutter/material.dart';
2 import 'home.dart';
3
4 void main() {
5   runApp(const MyApp());
6 }
7
8 class MyApp extends StatelessWidget {
9   const MyApp({super.key});
10
11   @override
12   Widget build(BuildContext context) {
13     return MaterialApp(
14       debugShowCheckedModeBanner: false,
15       title: 'Grade Calculator pro',
16       theme: ThemeData(
17         useMaterial3: true,
18         colorScheme: ColorScheme.fromSeed(
19           seedColor: Colors.teal,
20           primary: Colors.teal.shade700,
21         ), // ColorScheme.fromSeed
22         scaffoldBackgroundColor: Colors.grey.shade50,
23       ), // ThemeData
24       home: const MyHomePage(),
25     ); // MaterialApp
26   }
27 }
28
```

```
main.dart U home.dart 1, U X
1 import 'package:flutter/material.dart';
2 import 'result.dart';
3
4 enum Department { INE, INET }
5
6 class MyHomePage extends StatefulWidget {
7   const MyHomePage({super.key});
8   @override
9   State<MyHomePage> createState() => _MyHomePageState();
10 }
11
12 class _MyHomePageState extends State<MyHomePage> {
13   final _formKey = GlobalKey<FormState>();
14   final firstname = TextEditingController();
15   final lastname = TextEditingController();
16   final midScore = TextEditingController();
17   final finalScore = TextEditingController();
18   final classScore = TextEditingController();
19
20   Department? _department = Department.INE;
21   String? _selectedItem = "COMP NET DESIGN & IMPLEMENT";
22   List<String> dropdownmenu = [
23     "MOBILE APPLICATION DEVELOPMENT",
24     "FULL STACK SYSTEM DEVELOPMENT",
25     "COMP NET DESIGN & IMPLEMENT",
26     "DATABASE SYSTEM",
27   ];
28
29   @override
30   Widget build(BuildContext context) {
31     return Scaffold(
32       appBar: AppBar(
33         title: const Text(
34           'Academic Grading',
35           style: TextStyle(fontWeight: FontWeight.bold, color: Colors.white),
36         ), // Text
37         centerTitle: true,
38         backgroundColor: Colors.teal.shade700,
39         elevation: 4,
40       ), // AppBar
41       body: SingleChildScrollView(
42         padding: const EdgeInsets.symmetric(horizontal: 16, vertical: 24),
43
44         padding: const EdgeInsets.symmetric(horizontal: 16, vertical: 24),
45         child: Form(
46           key: _formKey,
47           child: Column(
48             children: [
49               _buildSectionCard(
50                 title: "Personal Information",
51                 child: Column(
52                   children: [
53                     _buildTextField(firstname, "First Name", Icons.badge),
54                     const SizedBox(height: 16),
55                     _buildTextField(
56                       lastname,
57                       "Last Name",
58                       Icons.badge_outlined,
59                     ),
60                   ],
61                 ),
62               const SizedBox(height: 20),
63               _buildSectionCard(
64                 title: "Academic Details",
65                 child: Column(
66                   crossAxisAlignment: CrossAxisAlignment.start,
67                   children: [
68                     const Text(
69                       "Select Department",
70                       style: TextStyle(color: Colors.grey),
71                     ),
72                     Row(
73                       children: [
74                         _buildRadio(Department.INE, "INE"),
75                         _buildRadio(Department.INET, "INET"),
76                       ],
77                     ),
78                     const Divider(height: 30),
79                     const Text(
80                       "Subject Name",
81                       style: TextStyle(color: Colors.grey),
82                     ),
83                     DropdownButtonFormField(
84                       value: _selectedItem,
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83 value: _selectedItem,  
84 isExpanded: true,  
85 decoration: const InputDecoration(  
86   border: OutlineInputBorder,  
87 ), // InputDecoration  
88 items: dropdownmenu  
89   .map(  
90     (e) => DropdownMenuItem(  
91       value: e,  
92       child: Text(  
93         e,  
94         style: const TextStyle(fontSize: 14),  
95       ), // Text  
96     ), // DropdownMenuItem  
97   )  
98   .toList(),  
99   onChanged: (val) =>  
100     setState(() => _selectedItem = val as String?),  
101   ), // DropdownButtonFormField  
102 ], // Column  
103 }, // Column  
104 },  
105 const SizedBox(height: 20),  
106 _buildSectionCard(  
107   title: "Assessment Scores",  
108   child: Column(  
109     children: [  
110       _buildScoreField(  
111         midScore,  
112         "Midterm Exam (20%)",  
113         Icons.edit_document,  
114       ),  
115       _buildScoreField(  
116         finalScore,  
117         "Final Exam (40%)",  
118         Icons.assignment,  
119       ),  
120       _buildScoreField(  
121         classScore,  
122         "Classwork (40%)",  
123         Icons.grade,  
124       ),  
125     ],  
126   ),  
127 ),  
128 ),  
129 ElevatedButton(  
130   style: ElevatedButton.styleFrom(  
131     minimumSize: const Size(double.infinity, 56),  
132     backgroundColor: Colors.teal.shade700,  
133     foregroundColor: Colors.white,  
134     shape: RoundedRectangleBorder(  
135       borderRadius: BorderRadius.circular(16),  
136     ),  
137     elevation: 5,  
138   ),  
139   onPressed: () {  
140     if (_formKey.currentState!.validate()) {  
141       Navigator.push(  
142         context,  
143         MaterialPageRoute(  
144           builder: (context) => MyResult(  
145             fname: firstname.text,  
146             lname: lastname.text,  
147             dept: department!.name,  
148             subject: _selectedItem!,  
149             midterm: midScore.text,  
150             finalScore: finalScore.text,  
151             classwork: classScore.text,  
152           ),  
153         ),  
154       );  
155     }  
156   },  
157   child: const Text(  
158     'CALCULATE RESULT',  
159     style: TextStyle(  
160       fontSize: 16,  
161       letterSpacing: 1.2,  
162       fontWeight: FontWeight.bold,  
163     ),  
164   ),  
165 ),  
166 ),
```

```

main.dart t.u X home.dart t.u X
165         ), // ElevatedButton
166         ],
167       ), // Column
168     ), // Form
169   ), // SingleChildScrollView
170 ); // Scaffold
171 }
172
173 Widget _buildSectionCard((required String title, required Widget child)) {
174   return Container(
175     padding: const EdgeInsets.all(20),
176     decoration: BoxDecoration(
177       color: Colors.white,
178       borderRadius: BorderRadius.circular(20),
179       boxShadow: [
180         BoxShadow(
181           color: Colors.black.withOpacity(0.05),
182           blurRadius: 10,
183           offset: const Offset(0, 4),
184         ), // BoxShadow
185       ],
186     ), // BoxDecoration
187     child: Column(
188       crossAxisAlignment: CrossAxisAlignment.start,
189       children: [
190         Text(
191           title,
192           style: TextStyle(
193             fontSize: 14,
194             fontWeight: FontWeight.bold,
195             color: Colors.teal.shade900,
196           ), // TextStyle
197         ), // Text
198         const SizedBox(height: 16),
199         child,
200       ],
201     ), // Column
202   ); // Container
203 }
204
205 Widget _buildTextField(
206   TextEditingController controller,
207   String label,
208   IconData icon,
209 ) {
210   return TextFormField(
211     controller: controller,
212     decoration: InputDecoration(
213       labelText: label,
214       prefixIcon: Icon(icon, color: Colors.teal),
215       filled: true,
216       fillColor: Colors.grey.shade50,
217     ),
218   );
219 }
220
221 Widget _buildRadio(Department val, String title) {
222   return Expanded(
223     child: RadioListTile(Department>{
224       value: val,
225       groupValue: _department,
226       title: Text(title, style: const TextStyle(fontSize: 14)),
227       contentPadding: EdgeInsets.zero,
228       onChanged: (v) => setState(() => _department = v),
229     },
230   );
231 }
232
233 Widget _buildScoreField(
234   TextEditingController controller,
235   String label,
236   IconData icon,
237 ) {
238   return Padding(
239     padding: const EdgeInsets.only(bottom: 12),
240     child: TextFormField(
241       controller: controller,
242       keyboardType: TextInputType.number,
243       decoration: InputDecoration(
244         labelText: label,
245         prefixIcon: Icon(icon, size: 20),
246         suffixText: "/ 100",
247         contentPadding: const EdgeInsets.symmetric(horizontal: 16),
248       ),
249       validator: (val) => val!.isEmpty ? 'Enter score' : null,
250     ),
251   );
252 }
253
254 Widget _buildRadio(Department val, String title) {
255   return Expanded(
256     child: RadioListTile(Department>{
257       value: val,
258       groupValue: _department,
259       title: Text(title, style: const TextStyle(fontSize: 14)),
260       contentPadding: EdgeInsets.zero,
261       onChanged: (v) => setState(() => _department = v),
262     },
263   );
264 }
265
266 Widget _buildRadio(Department val, String title) {
267   return Expanded(
268     child: RadioListTile(Department>{
269       value: val,
270       groupValue: _department,
271       title: Text(title, style: const TextStyle(fontSize: 14)),
272       contentPadding: EdgeInsets.zero,
273       onChanged: (v) => setState(() => _department = v),
274     },
275   );
276 }
277
278 Widget _buildRadio(Department val, String title) {
279   return Expanded(
280     child: RadioListTile(Department>{
281       value: val,
282       groupValue: _department,
283       title: Text(title, style: const TextStyle(fontSize: 14)),
284       contentPadding: EdgeInsets.zero,
285       onChanged: (v) => setState(() => _department = v),
286     },
287   );
288 }
289
290 Widget _buildRadio(Department val, String title) {
291   return Expanded(
292     child: RadioListTile(Department>{
293       value: val,
294       groupValue: _department,
295       title: Text(title, style: const TextStyle(fontSize: 14)),
296       contentPadding: EdgeInsets.zero,
297       onChanged: (v) => setState(() => _department = v),
298     },
299   );
300 }
301
302 Widget _buildRadio(Department val, String title) {
303   return Expanded(
304     child: RadioListTile(Department>{
305       value: val,
306       groupValue: _department,
307       title: Text(title, style: const TextStyle(fontSize: 14)),
308       contentPadding: EdgeInsets.zero,
309       onChanged: (v) => setState(() => _department = v),
310     },
311   );
312 }
313
314 Widget _buildRadio(Department val, String title) {
315   return Expanded(
316     child: RadioListTile(Department>{
317       value: val,
318       groupValue: _department,
319       title: Text(title, style: const TextStyle(fontSize: 14)),
320       contentPadding: EdgeInsets.zero,
321       onChanged: (v) => setState(() => _department = v),
322     },
323   );
324 }
325
326 Widget _buildRadio(Department val, String title) {
327   return Expanded(
328     child: RadioListTile(Department>{
329       value: val,
330       groupValue: _department,
331       title: Text(title, style: const TextStyle(fontSize: 14)),
332       contentPadding: EdgeInsets.zero,
333       onChanged: (v) => setState(() => _department = v),
334     },
335   );
336 }
337
338 Widget _buildRadio(Department val, String title) {
339   return Expanded(
340     child: RadioListTile(Department>{
341       value: val,
342       groupValue: _department,
343       title: Text(title, style: const TextStyle(fontSize: 14)),
344       contentPadding: EdgeInsets.zero,
345       onChanged: (v) => setState(() => _department = v),
346     },
347   );
348 }
349
350 Widget _buildRadio(Department val, String title) {
351   return Expanded(
352     child: RadioListTile(Department>{
353       value: val,
354       groupValue: _department,
355       title: Text(title, style: const TextStyle(fontSize: 14)),
356       contentPadding: EdgeInsets.zero,
357       onChanged: (v) => setState(() => _department = v),
358     },
359   );
360 }
361
362 Widget _buildRadio(Department val, String title) {
363   return Expanded(
364     child: RadioListTile(Department>{
365       value: val,
366       groupValue: _department,
367       title: Text(title, style: const TextStyle(fontSize: 14)),
368       contentPadding: EdgeInsets.zero,
369       onChanged: (v) => setState(() => _department = v),
370     },
371   );
372 }
373
374 Widget _buildRadio(Department val, String title) {
375   return Expanded(
376     child: RadioListTile(Department>{
377       value: val,
378       groupValue: _department,
379       title: Text(title, style: const TextStyle(fontSize: 14)),
380       contentPadding: EdgeInsets.zero,
381       onChanged: (v) => setState(() => _department = v),
382     },
383   );
384 }
385
386 Widget _buildRadio(Department val, String title) {
387   return Expanded(
388     child: RadioListTile(Department>{
389       value: val,
390       groupValue: _department,
391       title: Text(title, style: const TextStyle(fontSize: 14)),
392       contentPadding: EdgeInsets.zero,
393       onChanged: (v) => setState(() => _department = v),
394     },
395   );
396 }
397
398 Widget _buildRadio(Department val, String title) {
399   return Expanded(
400     child: RadioListTile(Department>{
401       value: val,
402       groupValue: _department,
403       title: Text(title, style: const TextStyle(fontSize: 14)),
404       contentPadding: EdgeInsets.zero,
405       onChanged: (v) => setState(() => _department = v),
406     },
407   );
408 }
409
410 Widget _buildRadio(Department val, String title) {
411   return Expanded(
412     child: RadioListTile(Department>{
413       value: val,
414       groupValue: _department,
415       title: Text(title, style: const TextStyle(fontSize: 14)),
416       contentPadding: EdgeInsets.zero,
417       onChanged: (v) => setState(() => _department = v),
418     },
419   );
420 }
421
422 Widget _buildRadio(Department val, String title) {
423   return Expanded(
424     child: RadioListTile(Department>{
425       value: val,
426       groupValue: _department,
427       title: Text(title, style: const TextStyle(fontSize: 14)),
428       contentPadding: EdgeInsets.zero,
429       onChanged: (v) => setState(() => _department = v),
430     },
431   );
432 }
433
434 Widget _buildRadio(Department val, String title) {
435   return Expanded(
436     child: RadioListTile(Department>{
437       value: val,
438       groupValue: _department,
439       title: Text(title, style: const TextStyle(fontSize: 14)),
440       contentPadding: EdgeInsets.zero,
441       onChanged: (v) => setState(() => _department = v),
442     },
443   );
444 }
445
446 Widget _buildRadio(Department val, String title) {
447   return Expanded(
448     child: RadioListTile(Department>{
449       value: val,
450       groupValue: _department,
451       title: Text(title, style: const TextStyle(fontSize: 14)),
452       contentPadding: EdgeInsets.zero,
453       onChanged: (v) => setState(() => _department = v),
454     },
455   );
456 }
457
458 Widget _buildRadio(Department val, String title) {
459   return Expanded(
460     child: RadioListTile(Department>{
461       value: val,
462       groupValue: _department,
463       title: Text(title, style: const TextStyle(fontSize: 14)),
464       contentPadding: EdgeInsets.zero,
465       onChanged: (v) => setState(() => _department = v),
466     },
467   );
468 }
469
470 Widget _buildRadio(Department val, String title) {
471   return Expanded(
472     child: RadioListTile(Department>{
473       value: val,
474       groupValue: _department,
475       title: Text(title, style: const TextStyle(fontSize: 14)),
476       contentPadding: EdgeInsets.zero,
477       onChanged: (v) => setState(() => _department = v),
478     },
479   );
480 }
481
482 Widget _buildRadio(Department val, String title) {
483   return Expanded(
484     child: RadioListTile(Department>{
485       value: val,
486       groupValue: _department,
487       title: Text(title, style: const TextStyle(fontSize: 14)),
488       contentPadding: EdgeInsets.zero,
489       onChanged: (v) => setState(() => _department = v),
490     },
491   );
492 }
493
494 Widget _buildRadio(Department val, String title) {
495   return Expanded(
496     child: RadioListTile(Department>{
497       value: val,
498       groupValue: _department,
499       title: Text(title, style: const TextStyle(fontSize: 14)),
500       contentPadding: EdgeInsets.zero,
501       onChanged: (v) => setState(() => _department = v),
502     },
503   );
504 }
505
506 Widget _buildRadio(Department val, String title) {
507   return Expanded(
508     child: RadioListTile(Department>{
509       value: val,
510       groupValue: _department,
511       title: Text(title, style: const TextStyle(fontSize: 14)),
512       contentPadding: EdgeInsets.zero,
513       onChanged: (v) => setState(() => _department = v),
514     },
515   );
516 }
517
518 Widget _buildRadio(Department val, String title) {
519   return Expanded(
520     child: RadioListTile(Department>{
521       value: val,
522       groupValue: _department,
523       title: Text(title, style: const TextStyle(fontSize: 14)),
524       contentPadding: EdgeInsets.zero,
525       onChanged: (v) => setState(() => _department = v),
526     },
527   );
528 }
529
530 Widget _buildRadio(Department val, String title) {
531   return Expanded(
532     child: RadioListTile(Department>{
533       value: val,
534       groupValue: _department,
535       title: Text(title, style: const TextStyle(fontSize: 14)),
536
```

grade

```
main.dart u | home.dart u | result.dart u x
1 import 'package:flutter/material.dart';
2
3 class MyResult extends StatelessWidget {
4   final String fname, lname, dept, subject, midterm, finalScore, classwork;
5   const MyResult({
6     super.key,
7     required this.fname,
8     required this.lname,
9     required this.dept,
10    required this.subject,
11    required this.midterm,
12    required this.finalScore,
13    required this.classwork,
14  });
15
16  @override
17  Widget build(BuildContext context) {
18    double m = double.tryParse(midterm) ?? 0;
19    double f = double.tryParse(finalScore) ?? 0;
20    double c = double.tryParse(classwork) ?? 0;
21    double total = m + f + c;
22
23    // Main
24    String grade = total >= 80
25      ? 'A'
26      : total >= 70
27      ? 'B'
28      : total >= 60
29      ? 'C'
30      : total >= 50
31      ? 'D'
32      : 'F';
33
34    Color gradeColor = grade == 'F'
35      ? Colors.red.shade700
36      : Colors.teal.shade700;
37
38    return Scaffold(
39      appBar: AppBar(
40        title: const Text('Result Summary'),
41        backgroundColor: Colors.teal.shade700,
42        foregroundColor: Colors.white,
43      ), // AppBar
44      body: Padding(
45        padding: const EdgeInsets.all(24.0),
46        child: Column(
47          children: [
48            // Main: Main
49            Container(
50              width: double.infinity,
51              padding: const EdgeInsets.all(24),
52              decoration: BoxDecoration(
53                gradient: LinearGradient(
54                  colors: [gradeColor, gradeColor.withOpacity(0.8)],
55                ),
56              borderRadius: BorderRadius.circular(24),
57            ),
58            child: Column(
59              children: [
60                Text(
61                  '$fname $lname',
62                  style: const TextStyle(
63                    color: Colors.white,
64                    fontSize: 22,
65                    fontWeight: FontWeight.bold,
66                  ),
67                ),
68                Text(dept, style: const TextStyle(color: Colors.white70)),
69                const SizedBox(height: 16),
70                CircleAvatar(
71                  radius: 50,
72                  backgroundColor: Colors.white,
73                  child: Text(
74                    grade,
75                    style: TextStyle(
76                      fontSize: 48,
77                      fontWeight: FontWeight.bold,
78                      color: gradeColor,
79                    ),
80                  ),
81                ),
82                const SizedBox(height: 12),
83                const Text(
84                  'FINAL GRADE',
85                  style: TextStyle(
86                    color: Colors.white,
87                    letterSpacing: 2,
88                    fontSize: 12,
89                  ),
90                ),
91              ],
92            ),
93            const SizedBox(height: 24),
94            // Final Score: Final Score
95            _buildInfoFile(Icons.book, "Subject", subject),
96            _buildInfoFile(
97              Icons.analytics,
98              "Total Score",
99              "${total.toStringAsFixed(1)} / 100",
100            ),
101            const Divider(height: 40),
102          ],
103        ),
104      ),
105    );
106  }
107
108  Widget _buildInfoFile(IconData icon, String label, String value) {
109    return ListTile(
110      leading: Icon(icon, color: Colors.teal),
111      title: Text(
112        label,
113        style: const TextStyle(fontSize: 12, color: Colors.grey),
114      ),
115      subtitle: Text(
116        value,
117        style: const TextStyle(fontSize: 16, fontWeight: FontWeight.bold),
118      ),
119    );
120  }
121
122  Widget _buildScoreStat(String label, String value) {
123    return Column(
124      children: [
125        Text(
126          value,
127          style: const TextStyle(fontSize: 20, fontWeight: FontWeight.bold),
128        ),
129        Text(label, style: const TextStyle(color: Colors.grey, fontSize: 12)),
130      ],
131    );
132  }
133}
```

```
main.dart u | home.dart u | result.dart u x
100
101 const Divider(height: 40),
102 Row(
103   mainAxisAlignment: MainAxisAlignment.spaceAround,
104   children: [
105     _buildScoreStat("Mid", midterm),
106     _buildScoreStat("Final", finalScore),
107     _buildScoreStat("Class", classwork),
108   ],
109 ), // Row
110 const Spacer(),
111 OutlinedButton(
112   onPressed: () => Navigator.pop(context),
113   style: OutlinedButton.styleFrom(
114     minimumSize: const Size(double.infinity, 56),
115     side: BorderSide(color: gradeColor),
116     shape: RoundedRectangleBorder(
117       borderRadius: BorderRadius.circular(16),
118     ), // RoundedRectangleBorder
119   ),
120   child: Text(
121     'BACK TO FORM',
122     style: TextStyle(
123       color: gradeColor,
124       fontWeight: FontWeight.bold,
125     ), // TextStyle
126   ), // Text
127 ), // OutlinedButton
128 ],
129 ), // Column
130 ), // Padding
131 ); // Scaffold
132
133
134 Widget _buildInfoFile(IconData icon, String label, String value) {
135   return ListTile(
136     leading: Icon(icon, color: Colors.teal),
137     title: Text(
138       label,
139       style: const TextStyle(fontSize: 12, color: Colors.grey),
140     ), // Text
141     subtitle: Text(
142       value,
143       style: const TextStyle(fontSize: 16, fontWeight: FontWeight.bold),
144     ), // Text
145   ); // ListTile
146 }
147
148 Widget _buildScoreStat(String label, String value) {
149   return Column(
150     children: [
151       Text(
152         value,
153         style: const TextStyle(fontSize: 20, fontWeight: FontWeight.bold),
154       ),
155       Text(label, style: const TextStyle(color: Colors.grey, fontSize: 12)),
156     ],
157   );
158 }
159
160
161 OutlinedButton(
162   onPressed: () => Navigator.pop(context),
163   style: OutlinedButton.styleFrom(
164     minimumSize: const Size(double.infinity, 56),
165     side: BorderSide(color: gradeColor),
166     shape: RoundedRectangleBorder(
167       borderRadius: BorderRadius.circular(16),
168     ),
169   ),
170   child: Text(
171     'BACK TO FORM',
172     style: TextStyle(
173       color: gradeColor,
174       fontWeight: FontWeight.bold,
175     ),
176   ),
177 ),
178 ],
179 ),
180 ],
181 );
182
183 Widget _buildInfoFile(IconData icon, String label, String value) {
184   return ListTile(
185     leading: Icon(icon, color: Colors.teal),
186     title: Text(
187       label,
188       style: const TextStyle(fontSize: 12, color: Colors.grey),
189     ),
190     subtitle: Text(
191       value,
192       style: const TextStyle(fontSize: 16, fontWeight: FontWeight.bold),
193     ),
194   );
195 }
196
197 Widget _buildScoreStat(String label, String value) {
198   return Column(
199     children: [
200       Text(
201         value,
202         style: const TextStyle(fontSize: 20, fontWeight: FontWeight.bold),
203       ),
204       Text(label, style: const TextStyle(color: Colors.grey, fontSize: 12)),
205     ],
206   );
207 }
```